# Hints: Combinatorics – 1. Counting and bijections

Homework

*The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.*

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## Problem 1

**Hint 1.** Identify the three colours with $0, 1, 2 \in \mathbb{Z}/3$. For three colours $a, b, c$, when is "two equal, one different" true, in terms of $a + b + c \bmod 3$?

**Hint 2.** All equal or all distinct is exactly $a + b + c \equiv 0$; so the condition is that every face sum is non-zero mod 3. Consider the linear map $\mathbb{F}_3^{30} \to \mathbb{F}_3^{20}$ sending an edge colouring to its 20 face sums; the answer is $2^{20}$ times the size of a fibre, provided the map is surjective.

**Hint 3.** The map is the unsigned vertex–edge incidence map of the dual graph (the dodecahedron: faces are vertices, edges join adjacent faces). For a connected non-bipartite graph this map is onto over a field of characteristic $\neq 2$ (adjust values along a spanning tree, and use an odd cycle to fix the last degree of freedom). The kernel has dimension 10, so the answer is $2^{20}3^{10}$.

## Problem 2

**Hint 1.** Note that the weights $1, 1, 2, \ldots, 63$ add up to 2017. Let $N_c$ be the number of tuples with weighted sum $\equiv c$. Multiplying all $x_i$ by a non-zero residue shows $N_c$ is the same for all $c \neq 0$, and $N_0 + 2016N_1 = 2017!/1953!$. One more relation is needed.

**Hint 2.** Compute $S = \sum \omega^{x_0 + x_1 + 2x_2 + \cdots + 63x_{63}}$ over tuples of distinct elements, with $\omega$ a primitive 2017th root of unity; then $S = N_0 - N_1$. Remove the distinctness condition by Möbius inversion over set partitions of the 64 indices: a partition $\pi$ contributes $\mu(\pi) \prod_{B \in \pi} \sum_x \omega^{w(B)x}$, where $w(B)$ is the total weight of block $B$.

**Hint 3.** An inner sum is 2017 if $w(B) \equiv 0$ and 0 otherwise, and only the whole set has weight divisible by 2017. So only the one-block partition survives, with $\mu = (-1)^{63}63!$, giving $N_0 - N_1 = -2017 \cdot 63!$. Solving: $N_0 = \frac{2016!}{1953!} - 2016 \cdot 63!$.

## Problem 3

**Hint 1.** Write $N = \frac{10^a - 1}{9} \cdot 10^b$ with $a \geqslant 1$ ones and $b \geqslant 0$ zeros, $a + b \leqslant 2020$. Factor $2020 = 4 \cdot 5 \cdot 101$.

**Hint 2.** $4 \cdot 5 \mid N$ iff $b \geqslant 2$, and $101 \mid N$ iff $10^a \equiv 1 \pmod{101}$ iff $4 \mid a$ (the order of 10 mod 101 is 4). Count the pairs $a = 4k$, $2 \leqslant b \leqslant 2020 - 4k$: $\sum_{k=1}^{504}(2019 - 4k) = 508536$.

## Problem 4

**Hint 1.** Use the Chinese remainder theorem: $x^2 - x = x(x - 1)$, and $x$, $x - 1$ are coprime.

**Hint 2.** $10^{2006} \mid x(x - 1)$ iff $x \equiv 0$ or $1 \bmod 2^{2006}$ and $x \equiv 0$ or $1 \bmod 5^{2006}$: four residues mod $10^{2006}$, one of which is 0. The answer is 3.

## Problem 5

**Hint 1.** Factor over $\mathbb{C}$: $(P + iQ)(P - iQ) = X^{2n} + 1$. What are the degrees and leading coefficients of the two factors, given $\deg Q < \deg P$?

**Hint 2.** $\deg P = n$, and $P + iQ$ has real leading coefficient $c$ with $c^2 = 1$. Its roots are $n$ of the $2n$ roots of $X^{2n} = -1$, and $P - iQ$ (its conjugate) takes the conjugate roots, so $P + iQ$ contains exactly one root from each conjugate pair. Conversely each choice gives valid real $P, Q$ with $\deg Q < n$. The answer is $2 \cdot 2^n = 2^{n+1}$.

## Problem 6

**Hint 1.** Try small grids ($2 \times n$, $3 \times 3$). Each coin moves along its diagonal; describe which sets of coins have moved (each coin moves at most once, since afterwards some square it needs is occupied).

**Hint 2.** Show by induction on moves that in every reachable configuration the moved coins form an "up-right closed" set: there is a monotone lattice path from the bottom-left to the top-right corner of the $(m-1) \times (n-1)$ board separating moved coins from unmoved ones, and conversely every such path is reachable. The answer is $\binom{m+n-2}{m-1}$.

## Problem 7

*Note.* as stated the answer is infinite. The original (Putnam 2011 A1) asks how many points $(x, y)$ with $0 \leqslant x, y \leqslant 2011$ are **not** the endpoint of a growing spiral; hints are for that version.

**Hint 1.** With lengths $a_1 < a_2 < \ldots$, the endpoint is $x = a_1 - a_3 + a_5 - \ldots$, $y = a_2 - a_4 + \ldots$. For an endpoint in the closed first quadrant, determine which $n \pmod 4$ are possible by looking at the sign of the last horizontal and vertical pairs.

**Hint 2.** Only $n = 2$ and $n \equiv 2 \pmod 4$, $n \geqslant 6$, give first-quadrant endpoints. With $n = 2$ you get exactly the points with $0 < x < y$. With $n \geqslant 6$ you get $x = a_1 + (a_5 - a_3) + \ldots \geqslant 3$ and $y = a_2 + (a_6 - a_4) + \ldots \geqslant 4$, and $n = 6$ already realises every such point.

**Hint 3.** So the non-endpoints are: $x = 0$ (2012 points), $(1, y)$ with $y \leqslant 1$, $(2, y)$ with $y \leqslant 2$, and $(x, y)$ with $x \geqslant 3$, $y \leqslant 3$. Total $2012 + 2 + 3 + 4 \cdot 2009 = 10053$.

## Problem 8

**Hint 1.** Only residues mod 3 matter. Among $1, \ldots, 3k+1$ there are $k+1$ numbers $\equiv 1$, $k$ numbers $\equiv 2$, $k$ numbers $\equiv 0$. Zeros do not change the running sum mod 3 (but cannot come first).

**Hint 2.** Ignoring zeros, the running sum must avoid 0 mod 3, which forces the residue sequence $1, 1, 2, 1, 2, \ldots, 1, 2$ (or starting $2, 2, 1, \ldots$, impossible with these counts). Compute the probability that the non-zero residues come in this order and that the first number is not $\equiv 0$; the answer is $\frac{k!\,(k+1)!}{(3k+1)\,(2k)!}$.

## Problem 9

**Hint 1.** Inclusion–exclusion with $A_i =$ maps $[n] \to [k]$ missing the value $i$; $\left|\bigcap_{i \in I} A_i\right| = (k - |I|)^n$.

**Hint 2.** A surjection is the same as a partition of $[n]$ into $k$ non-empty blocks together with a bijection from blocks to $[k]$; hence the count is $k!\,S(n,k)$ and $S(n,k) = \frac{1}{k!} \sum_j (-1)^j \binom{k}{j} (k-j)^n$.

## Problem 10

**Hint 1.** $f(n)$ counts tuples $(c_1, \ldots, c_n)$ of non-negative integers with $\sum c_k\, k! = n!$. Once $c_2, \ldots, c_n$ are chosen, $c_1$ is determined (if non-negative). Both bounds come from estimating $\prod_{k=1}^{n-1} \frac{n!}{k!}$.

**Hint 2.** Upper bound: $c_k \leqslant n!/k!$, so $f(n) \leqslant \prod_{k=1}^{n} (n!/k! + 1) \leqslant 2^n \prod_{k=1}^{n} n!/k!$. Lower bound: choose $c_k \in [0, \frac{n!}{n \cdot k!}]$ freely for $2 \leqslant k \leqslant n-1$ and $c_n = 0$; then the sum is at most $n!$ and $c_1$ completes it. This gives $f(n) \geqslant \prod_{k=2}^{n-1} \frac{n!}{n\,k!}$.

**Hint 3.** Use Stirling in the form $\log k! = k \log k - k + O(\log k)$: $\log \prod_{k \leqslant n} \frac{n!}{k!} = n \log n! - \sum_{k \leqslant n} \log k! = \frac{n^2}{2} \log n - \frac{n^2}{4} + O(n \log n)$. All the losses ($2^n$, $n^n$, the $O(n \log n)$ error) are of the form $n^{O(n)}$, which is what the constant $C$ absorbs.